1.2 Contributors to Microbiology

Many scientists contributed to the science of Microbiology from the 17th century to the present day. Some prominent microbiologists who have made significant contribution to the study of microorganisms are given below:

1.2.1 Antony Van Leeuwenhoek

Antony Van Leeuwenhoek (1632-1723) of Holland (Figure 1.1) developed microscopes. He was a Dutch merchant and a skilled lens maker. He made a variety of lenses with magnifying power 50-300X.

He was the first person to invent simple microscope. It has a single biconvex lens with a magnification of about 200X (Figure 1.2). His microscopes resolved bodies with diameters measuring below 1micron. He examined water, mud, saliva and found living organisms. He called these microorganisms as **Animalcules** (little animals). Bacteria like cocci, bacilli and spirochetes were recognized. He proposed that the size of bacteria is one sixth the diameter of Red Blood Cells.

He observed the growth of bacteria in infusions. The existence of spermatozoa and RBC was revealed by him. Animal histology was established by him. He described capillary circulation and added a new dimension to Biology. All kinds of unicellular microorganisms were accurately described by him including human oral microbial flora. He is commonly known as the '**Father of Microbiology**'.

1.2.2 Louis Pasteur (1822-1895)

Louis Pasteur was a French chemist and a crystallographer (Figure 1.3). His greatest contribution to microbiology made him to be the '**Father of Modern Microbiology**'.



Figure 1.1:
Antony Van Leeuwenhoek

Antony Van Leeuwenhoek wrote many letters. He wrote them in Dutch, the only language that he knew. These letters, described his complete scientific output. Antony Van Leeuwenhoek in a letter dated 12th June 1716, wrote "... *my work, which I've done for a long time, was not pursued in order to gain the praise I now enjoy, but chiefly from a craving after knowledge, which I notice resides in me more than in most other men. And therewithal, whenever I found out anything remarkable, I have thought it my duty to put down my discovery on paper, so that all ingenious people might be informed thereof*".

Contribution to science as a chemist

Louis Pasteur was working with tartaric acid crystals. He could pick up the dextro and levo rotatory crystals by seeing the morphology of the crystals. Later he was called to solve some of the problems in fermentation industry and turned his attention to biological process of fermentation.

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